



Personal Productivity App PRD

1. Product Overview

Product Name:

FlowSpace (placeholder name — can be changed later)

Goal / Vision:

To build a lightweight productivity app that allows individuals and small teams to structure their tasks and projects in a flexible, relational system — with visibility across all workspaces (Spaces, Lists, Tasks, and Projects). The goal is to create a manageable medium for daily task tracking and project progress visualization.

Target Users:

Individual freelancers, students, or small teams who want a clean, customizable, and affordable productivity solution.

Problem Statement:

Tools like ClickUp provide powerful task and project management features but enforce rigid hierarchy and limited relational visibility between entities. For example, users can link tasks to projects, but cannot display or aggregate those relationships across lists or spaces.

This leads to fragmented visibility — users can't see how tasks and projects relate in one unified view.

The solution is to create a streamlined app that enables fully visible relationships between spaces, lists, projects, and tasks — all accessible through simple, consistent views (Kanban, List, Calendar).

2. Product Goals & Success Metrics

Goal	Metric / KPI
Flexible relational linking between Tasks and Projects	Tasks show related Project in all views
Simplified daily task management	Task creation or update in <5 clicks
Consistent multi-view experience	Kanban, List, and Calendar show same dataset

Goal	Metric / KPI
Lightweight performance	Page load < 2 seconds
Self-hostable and affordable	No paid dependencies required for MVP

3. Core Modules

Module	Description	Priority
Spaces	Top-level work area grouping lists (e.g., Work, Personal)	High
Lists	Contain either Tasks or Projects within a space	High
Tasks	Atomic units of work with progress tracking	High
Projects	Larger work items that group multiple tasks	High
Views	Kanban, List, Calendar for visualization	High
Relationships	Manage links between Tasks and Projects	High
Settings	Workspace customization (theme, preferences)	Medium

4. Detailed Features

4.1 Spaces

Feature	Description	Notes
Create / Edit / Delete Space	Manage high-level work areas	Each space holds multiple lists
View All Lists in a Space	Display both task and project lists together	
Color / Icon Customization	Optional for visual grouping	v2 feature

4.2 Lists

Feature	Description	Notes
Create / Edit / Delete List	Manage categorized lists inside a space	Each list can either contain tasks or projects
Define List Type	Choose between “Task List” or “Project List”	Determines default fields and statuses
Custom Status Pipelines	Editable columns (Kanban) per list type	e.g., Task: To Do → In Progress → Done; Project: Planning → Pending Payment

4.3 Tasks

Feature	Description	Notes
CRUD Operations	Add, update, delete, or view tasks	Core function
Fields	Title, Description, Due Date, Priority, Status, Linked Project	
Kanban Drag & Drop	Move tasks across status columns	Real-time updates via Supabase
Relationship Display	Each task shows its linked project (if any)	Visible in all views
Calendar Integration	Display tasks by due date	Local calendar only for MVP
Quick Add Task	Inline creation within Kanban/List	

4.4 Projects

Feature	Description	Notes
CRUD Operations	Create, update, delete, view projects	Core function
Fields	Name, Description, Start Date, Status, Linked Tasks	
Project Status	Pipeline: Planning → In Progress → Pending Payment	
Progress Indicator	$\% \text{ Complete} = (\# \text{ Done tasks} / \text{Total tasks}) \times 100$	Auto-calculated
Relationship	Each project can have multiple tasks	One-to-many relationship

4.5 Views

View	Description	Core Functions
Kanban View	Column-based workflow visualization	Drag-and-drop between statuses
List View	Table layout for sorting and filtering	Filter by project, due date, or status
Calendar View	Date-based scheduling	Shows due/start dates for both tasks and projects

4.6 Relationships

Relationship	Type	Description
Space → List	1 → Many	A Space holds multiple Lists
List → Task	1 → Many	A Task List contains multiple Tasks
List → Project	1 → Many	A Project List contains multiple Projects
Project ↔ Task	1 → Many	A Project can have many Tasks; a Task belongs to one Project

5. Data Model (High-Level)

Entities:

- Space
- List
- Project
- Task

```
Space:
  id: UUID
  name: string
  description: string (nullable)
  created_at: datetime
```

```
List:
  id: UUID
  space_id: FK -> Space.id
  name: string
  type: enum('task', 'project')
  status_columns: json (array of status names)
  created_at: datetime
```

```
Project:
  id: UUID
  list_id: FK -> List.id
  name: string
  description: text
  start_date: date
  status: string
  progress: float (calculated)
  created_at: datetime
```

```
Task:
  id: UUID
  list_id: FK -> List.id
  project_id: FK -> Project.id (nullable)
  title: string
  description: text
  due_date: date (nullable)
  priority: enum('low', 'medium', 'high')
  status: string
  created_at: datetime
  updated_at: datetime
```

6. Non-Functional Requirements

Category	Requirement
Platform	Web app (Next.js + Supabase)
Performance	Page load < 2s
Security	Supabase Auth (email/password)
UX	Responsive layout, drag-and-drop Kanban, minimal UI
Scalability	Up to 10,000 tasks per user
Offline Handling	Local cache + sync (v2 feature)
Accessibility	Keyboard shortcuts, dark mode (v2)

7. MVP Scope (v1.0)

Feature	Included	Notes
Spaces	✓	Basic CRUD
Lists	✓	Task/Project type selection
Tasks	✓	CRUD + Kanban movement
Projects	✓	CRUD + linked tasks
Views	✓	Kanban, List, Calendar
Auth	✓	Email/password via Supabase
Relationships	✓	Task ↔ Project linkage
Progress Calculation	✓	Auto-updated from task completion
Dashboard	✗	Future enhancement

8. Future Enhancements (v2+)

- Team collaboration (multi-user sharing)
 - Task comments and attachments
 - Time tracking (per task)
 - AI task summaries and smart prioritization
 - Calendar sync (Google, Outlook)
 - Public project sharing for clients
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9. Technical Notes

Area	Recommendation
Frontend Framework	Next.js (React, TailwindCSS, Zustand for state)
Backend / DB	Supabase (PostgreSQL + Auth + Realtime)
Hosting	Supabase for backend, Vercel for frontend
State Management	Zustand or React Query
Drag & Drop	dnd-kit or react-beautiful-dnd
Calendar	react-big-calendar or fullcalendar-react
UI Components	shadcn/ui + TailwindCSS

Summary:

FlowSpace is a minimal, relational productivity system for individuals who need structure between tasks and projects, using a familiar ClickUp-like hierarchy — but with better relational visibility and simplicity.